

n generator) is rms. There are two types of oscillators : sinusoidal and non-sinusoidal. The latter employ non-linear elements in the feedback circuit.

OSCILLATORS

For a frequency-oscillator (a transistor or op-amp oscillator), the frequency at which oscillation occurs is determined by the feedback network, provided the loop gain is equal to, or greater than, unity.

We have already discussed the basic principle of an oscillator. The following circuit will generate a sinusoidal waveform.

The circuit generates a sinusoidal waveform using an op-amp which provides a phase shift of 180° as it is an inverting amplifier. The feedback network provides an additional phase shift of 180° (or 0°), so that the total phase shift is 360° (or 0°). The amplitude of the oscillation is determined by the feedback network, as shown in Fig. 31.1(a).

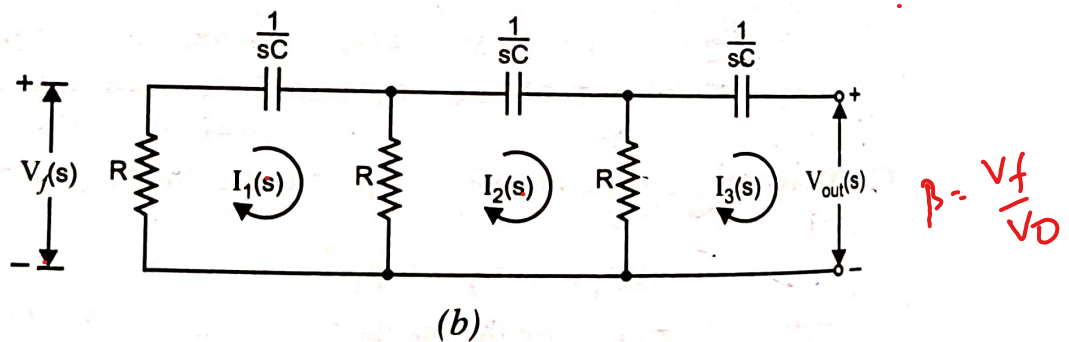
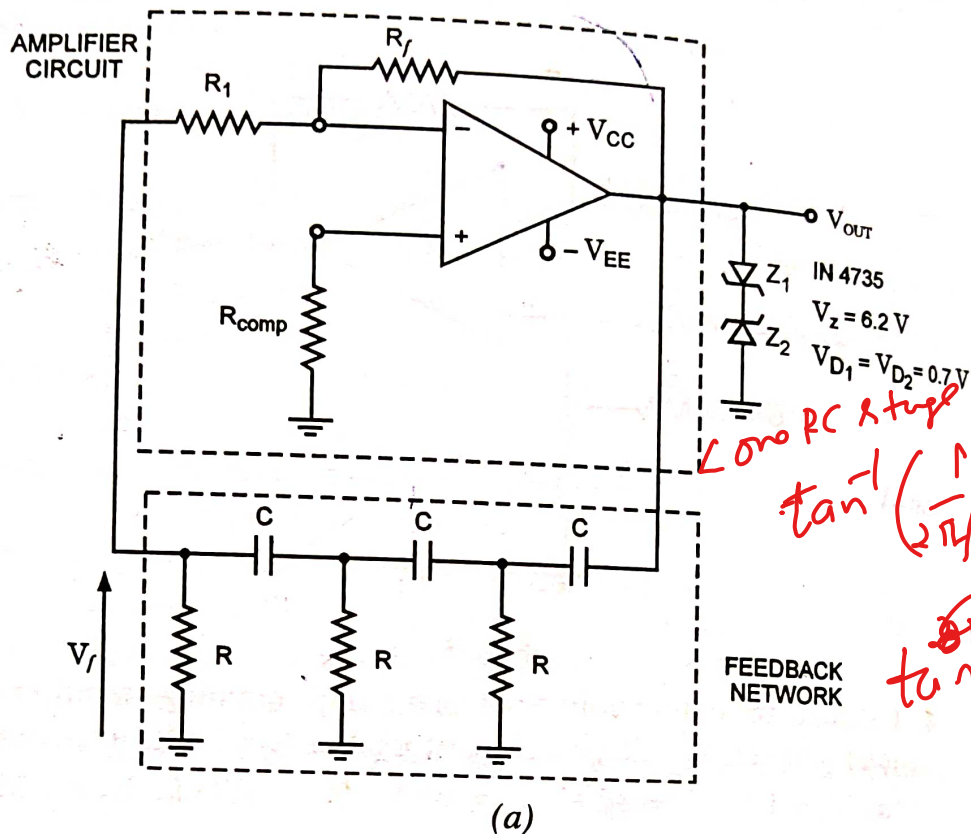


Fig. 31.1

$$0 = -RI_1(s) + \left(2R + \frac{1}{sC} \right) I_2(s) - RI_3(s) \quad \dots[31.1(b)]$$

$$-V_{out}(s) = -RI_2(s) + \left(R + \frac{1}{sC} \right) I_3(s) \quad \dots[31.1(c)]$$

In matrix form :

$$\begin{bmatrix} \hat{V}_f(s) \\ 0 \\ -V_{out}(s) \end{bmatrix} = \begin{bmatrix} R + \frac{1}{sC} & -R & 0 \\ -R & 2R + \frac{1}{sC} & -R \\ 0 & -R & R + \frac{1}{sC} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} \quad \dots(31.2)$$

...[31.1(a)]

$$I_1(s) = \frac{\begin{bmatrix} V_f(s) & -R & 0 \\ 0 & 2R + \frac{1}{sC} & -R \\ -V_{out}(s) & -R & R + \frac{1}{sC} \end{bmatrix}}{\Delta}$$

$$= \frac{V_f(s) \left(R^2 + \frac{3R}{sC} + \frac{1}{s^2 C^2} \right) - V_{out}(s) \cdot R^2}{\Delta} \dots(31.3)$$

$$\text{where } \Delta = \begin{bmatrix} R + \frac{1}{sC} & -R & 0 \\ -R & 2R + \frac{1}{sC} & -R \\ 0 & -R & R + \frac{1}{sC} \end{bmatrix}$$

$$= \frac{3R^2}{sC} + \frac{4R}{s^2 C^2} + \frac{1}{s^3 C^3} \dots(31.4)$$

Now from Fig. 31.1(b)

$$-V_f(s) = I_1(s)R \dots(31.5)$$

$$\text{or } -V_f(s) = \frac{\left[V_f(s) \left(R^2 + \frac{3R}{sC} + \frac{1}{s^2 C^2} \right) - V_{out}(s) R^2 \right] R}{\Delta}$$

$$\text{or } -V_f(s)\Delta = V_f(s) \left(R^3 + \frac{3R^2}{sC} + \frac{R}{s^2 C^2} \right) - V_{out}(s) R^3$$

$$\text{or } V_f(s) \left[R^3 + \frac{3R^2}{sC} + \frac{R}{s^2 C^2} + \Delta \right] = V_{out}(s) R^3$$

Substituting the value of Δ from Eq. (31.4) in above equation, we have

$$\text{or } V_f(s) \left[R^3 + \frac{3R^2}{sC} + \frac{R}{s^2 C^2} + \frac{3R^2}{sC} + \frac{4R}{s^2 C^2} + \frac{1}{s^3 C^3} \right] = V_{out}(s) R^3$$

$$\text{or } \frac{V_f(s)}{V_{out}(s)} = \frac{R^3}{R^3 + \frac{6R^2}{sC} + \frac{5R}{s^2 C^2} + \frac{1}{s^3 C^3}} \dots(31.6)$$

or feedback gain,

$$\beta = \frac{V_f(s)}{V_{out}(s)}$$

$$= \frac{1}{1 + \frac{6}{sCR} + \frac{5}{s^2 C^2 R^2} + \frac{1}{s^3 C^3 R^3}}$$

Substituting $s = j\omega$ in Eq. (31.7), we have

$$\beta = \frac{1}{1 + \frac{6}{j\omega CR} + \frac{5}{(j\omega CR)^2} + \frac{1}{(j\omega CR)^3}}$$

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Example 31
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... (31.7) Solution:

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$$= \frac{1}{1 - \frac{5}{(\omega CR)^2} - j \left[\frac{6}{\omega CR} - \frac{1}{(\omega CR)^3} \right]} \quad \dots(31.8)$$

To satisfy frequency of oscillation criteria, equate the imaginary components of Eq. (31.8) to zero (so that $A\beta = 1$ for which β must be real quantity).

$$\text{Hence } \frac{6}{\omega CR} - \frac{1}{(\omega CR)^3} = 0$$

$$\text{or } \frac{1}{\omega CR} \left(6 - \frac{1}{\omega^2 C^2 R^2} \right) = 0$$

$$\text{or } \frac{1}{\omega CR} = \sqrt{6}$$

$$\text{or } f = \frac{1}{2\pi CR\sqrt{6}} \quad \dots(31.9)$$

This is frequency of oscillation.

Also, the loop gain, $\beta A = -1$ (Barkhausen criterion)

$$\text{or } \beta A = 1 \angle 180^\circ$$

$$\text{or } \frac{1}{\left[1 - \frac{5}{(2\pi f CR)^2} \right]} A = 1 \angle 180^\circ$$

$$\text{or } \frac{1}{[1 - 5(\sqrt{6})^2]} A = 1 \angle 180^\circ$$

$$\text{or } A \geq -29$$

$\dots(31.10)$

i.e., the inverting amplifier should have gain of at least 29.

$$\text{or say } \left| \frac{R_f}{R_1} \right| = 29 \quad \dots[31.11(a)]$$

$$\text{i.e., } R_f = 29 R_1. \quad \dots[31.11(b)]$$

Thus, the circuit will produce a sinusoidal waveform of frequency f if the gain is at least 29 and total phase shift of the circuit is 360° .

The gain is kept greater than 29 so that any variation in circuit parameters cannot make $|A\beta| < 1$. Otherwise oscillations will die out.

$$f = \frac{0.065}{RC}$$

min gain of the amp^s to produce sustained oscillation.